

Intercontinental Transmission of Low Pathogenic Avian Influenza between Eurasia and North America (2002-2024): the Crucial Role of *Anseriformes* and *Charadriiformes*.

Ji-Yun Kim, Da-Won Kim, Dong-Wook Lee, Ye-Jin Lee and Jung-Hoon Kwon*
Laboratory of Veterinary Microbiology, College of Veterinary Medicine, Kyungpook National University; Daegu, 41566, Republic of Korea
E-mail address: skykjyun@naver.com



Introduction

- Avian influenza viruses (AIVs) are phylogenetically categorized into Eurasian and North American lineages according to geographical barriers
- Nevertheless, some migratory bird, particularly *Anseriformes* migrating across the Bering Sea such as *Anas acuta* and *Anser albifrons*, facilitates seasonal viral exchange between continents.
- Additionally, *Charadriiformes* also expected to contribute to intercontinental transmission due to their wide ecological distribution and extensive migratory behaviors.
- To gain comprehensive insight into the intercontinental movement of AIVs, we conducted the phylogenetic analysis on AIVs isolated from 2002 to 2024 and quantified patterns of intercontinental viral movement.
- We then organized the isolation region, host species, and subtype information for each virus associated with intercontinental transmission.
- Finally, we performed correlation analysis between temperature and intercontinental movement patterns to explore potential climate-related drivers.

Materials & Methods

- Preparing data set for phylogenetic analysis**
 - AIV sequences were retrieved from the GISAID (January 1, 2002-March 29, 2024) excluding highly pathogenic avian influenza H5 viruses.
 - The dataset comprised 69,966 Eurasian and 47,121 North American strains.
- Intercontinental viral transmission detection with phylogenetic analysis.**
 - A FastTree-based approximately-maximum-likelihood phylogeny was generated using 14,813 high-quality internal gene sequences.
 - Intercontinental transmission events were inferred via phylogenetic clustering discordant with isolated location.
 - The reassorted viruses with different subtype backbones were considered distinct introduction.
 - Even among virus sharing the same subtype backbone, those with genetic identity of less than 99% per year were also assumed to be distinct viral introductions.
 - For sequences with >99% identity/year, the earliest isolate was designated as the index case.
- Metadata analysis of intercontinental viral movement.**
 - Metadata (isolation date, host, location, subtype) were analyzed per transmission event.
 - Trends included annual transmission frequencies, host/subtype distributions by region, directional subtype variation, and seasonal occurrence patterns.
- Statistical analysis of factors influencing intercontinental viral transmission.**
 - Alaska and Iceland, critical migratory bird breeding sites, were analyzed for temperature-associated effects on transmission.
 - Correlation analyses (GraphPad Prism v10.3.1) included:
 - Annual isolate counts vs. transmission frequency
 - Mean annual/breeding season temperatures in Alaska and Iceland vs. transmission frequency/rate
 - Only years with ≥300 isolates were included in the temperature association analysis.

Result & Discussion

- According to the phylogenetic analyses, H13 and H16 subtypes formed separate monophyletic clusters (**Figure 1**).
- This demonstrates the H13 and H16 subtype undergo independent evolution of other LPAI subtypes, having adapted for replication and transmission in gulls.
- We identified 440 initial intercontinental AIV introductions. Intercontinental viral transmission was most frequent in 2015 (**Figure 2**).
- Intercontinental viral transmission events have been identified between Alaska and East Asia, as well as between Eastern North America and Western Europe.
- While *Anatidae* were the predominant hosts in intercontinental viral transmission events between North America and Eurasia, *Scolopacidae* were the most frequently isolated hosts in Eastern North America, whereas *Laridae* were predominant in Europe (**Figure 3**).
- H3N8 was the most frequently detected intercontinental reassortant subtype. *Anatidae* had the greatest subtype diversity, followed by *Scolopacidae* and *Laridae* (**Figure 4**).
- These findings indicate that both Beringia region and Atlantic flyway serve as critical hubs for intercontinental AIV transmission. In both regions, *Anseriformes* and *Charadriiformes* acts as principal host facilitating viral transmission.
- For Asia–North America transmission, peaks occurred in September; for Europe–North America, May was peak months, showing differences in directional peak timing. *Anatidae* dominated in Asia–North America, while *Scolopacidae* (spring) and *Anatidae* (late summer–winter) led in Europe–North America. *Laridae* peaked in winter (**Figure 5**).
- This reflects AIV ecology, with naïve juvenile waterfowl infected before or during fall migration. It supports prior findings of spring AIV peaks in Delaware Bay shorebirds and fall increases in *Laridae*. These patterns underscore host ecology's role in intercontinental AIV dynamics.
- No significant link was found between climate factors and intercontinental transmission in Alaska and Iceland. However, viral transmission count was positively correlated with total virus isolations ($p=0.001$) (**Figure 6**).
- Despite these findings, numerous studies have highlighted the influence of climate change on bird habitat and migration patterns. Therefore, the impact of climate change on AIV ecology should be estimated on a regular basis.
- Our study demonstrates a positive correlation between intercontinental viral transmission and viral isolations, underscoring the need for continued AIV surveillance. Inclusion of *Charadriiformes* in monitoring programs, especially in regions like South Korea, is essential for capturing the full scope of intercontinental AIV spread.

Result

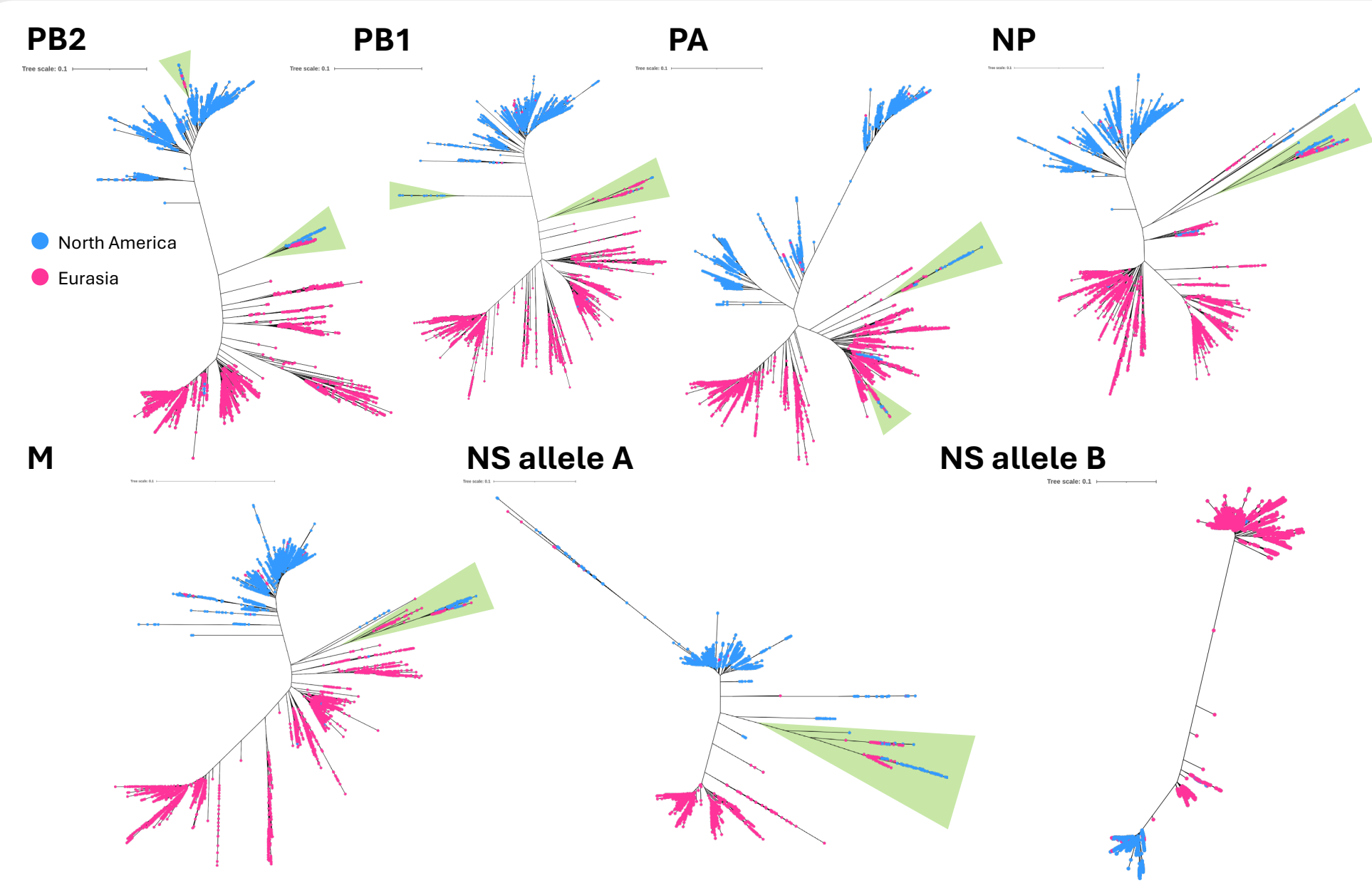


Figure 1. Approximately-maximum-likelihood tree of 6 internal gene segments.

The phylogenetic tree of 6 internal gene segments were visualized in unrooted tree layout. For the NS gene, sequences were visualized separately by allele A and allele B. AIVs isolated in Eurasia were indicated in red circles and AIVs isolated in North America were indicated in blue circles. Subtype H13 and H16 clusters were colored in green triangles.

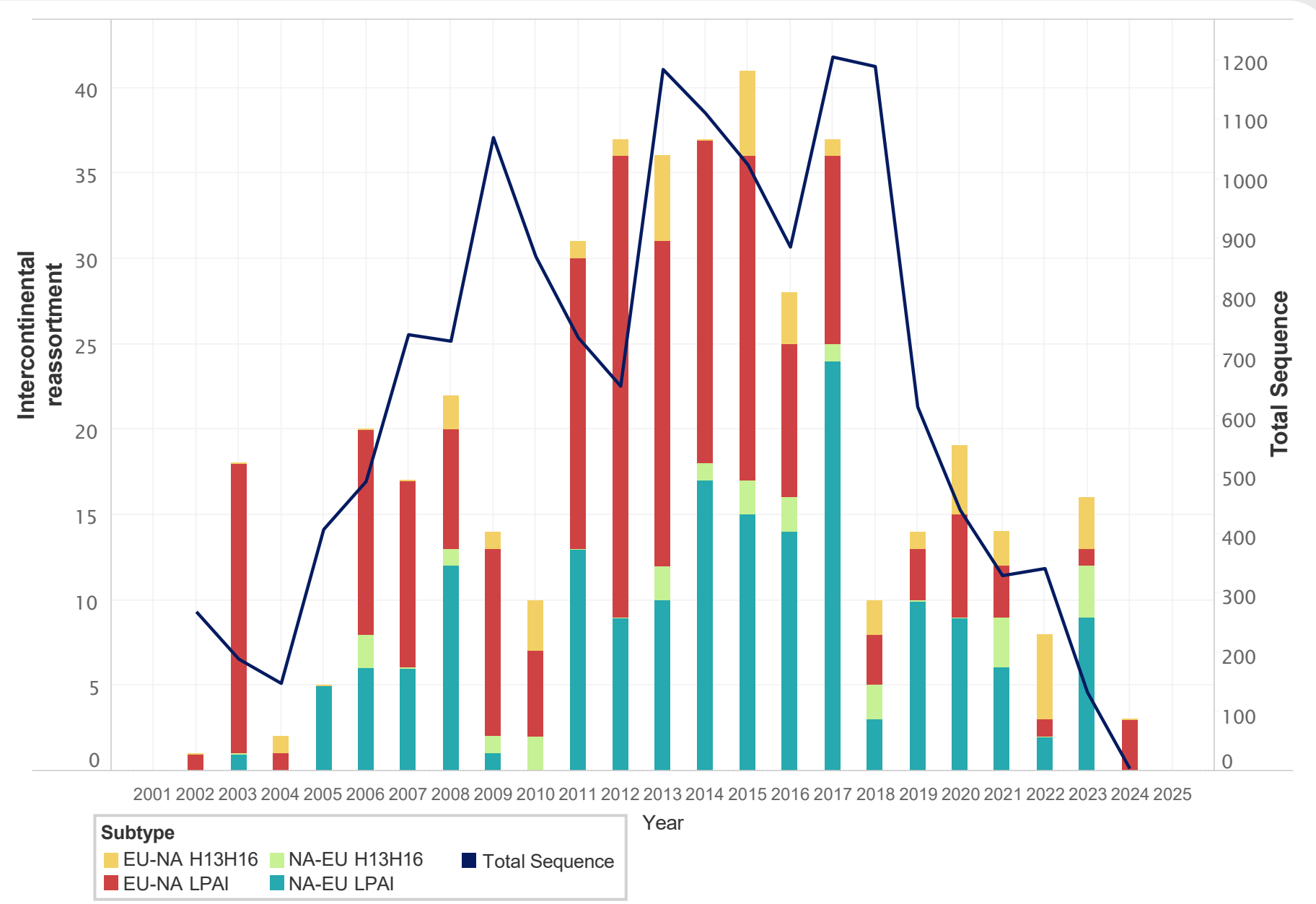


Figure 2. Trends in AIV Initial Intercontinental transmission and Total AIV isolates by Year

stacked bar charts show the number of intercontinental transmission event from North America to Eurasia (NA-EU) and from Eurasia to North America (EU-NA), categorized by subtype LPAI (HA subtypes excluding H5, H13 and H16) and H13, and H16. The total number of sequences analyzed per year was represented by the blue line.

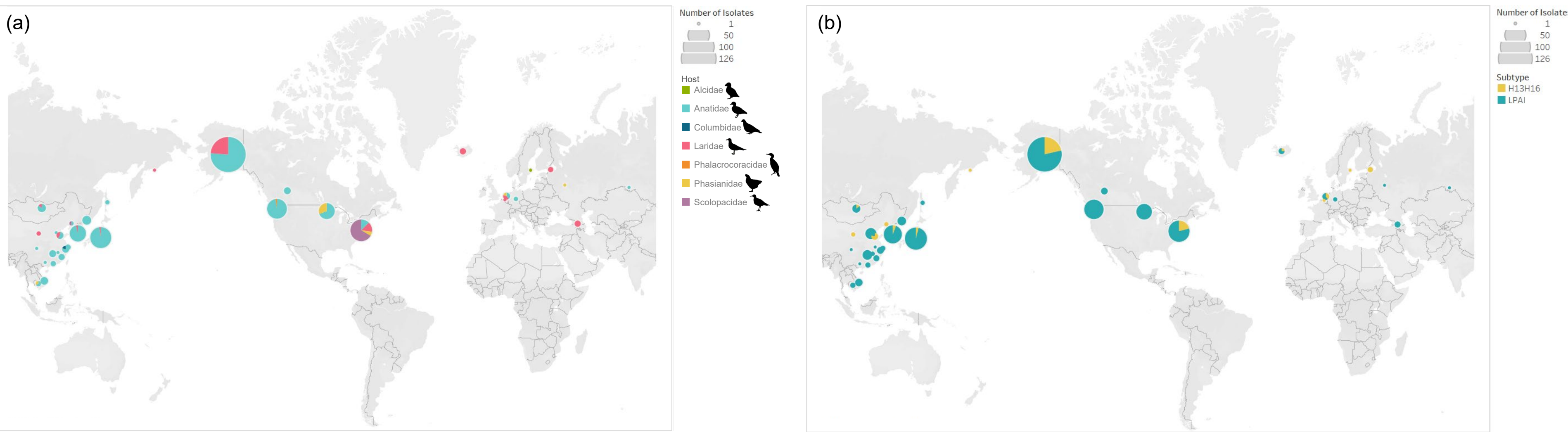


Figure 3. Global Distribution of Intercontinentally transmitted Avian Influenza Virus Isolates by Host and Subtype

the global distribution of avian influenza virus isolates that have undergone intercontinental transmission, highlighting data from two perspectives: host species (a) and virus subtype (b). (a) The geographic locations where avian influenza viruses were isolated, color-coded by host species. (b) The geographic locations, color-coded by virus subtypes, specifically H13, H16 and LPAI. The size of each circle represents the number of isolates.

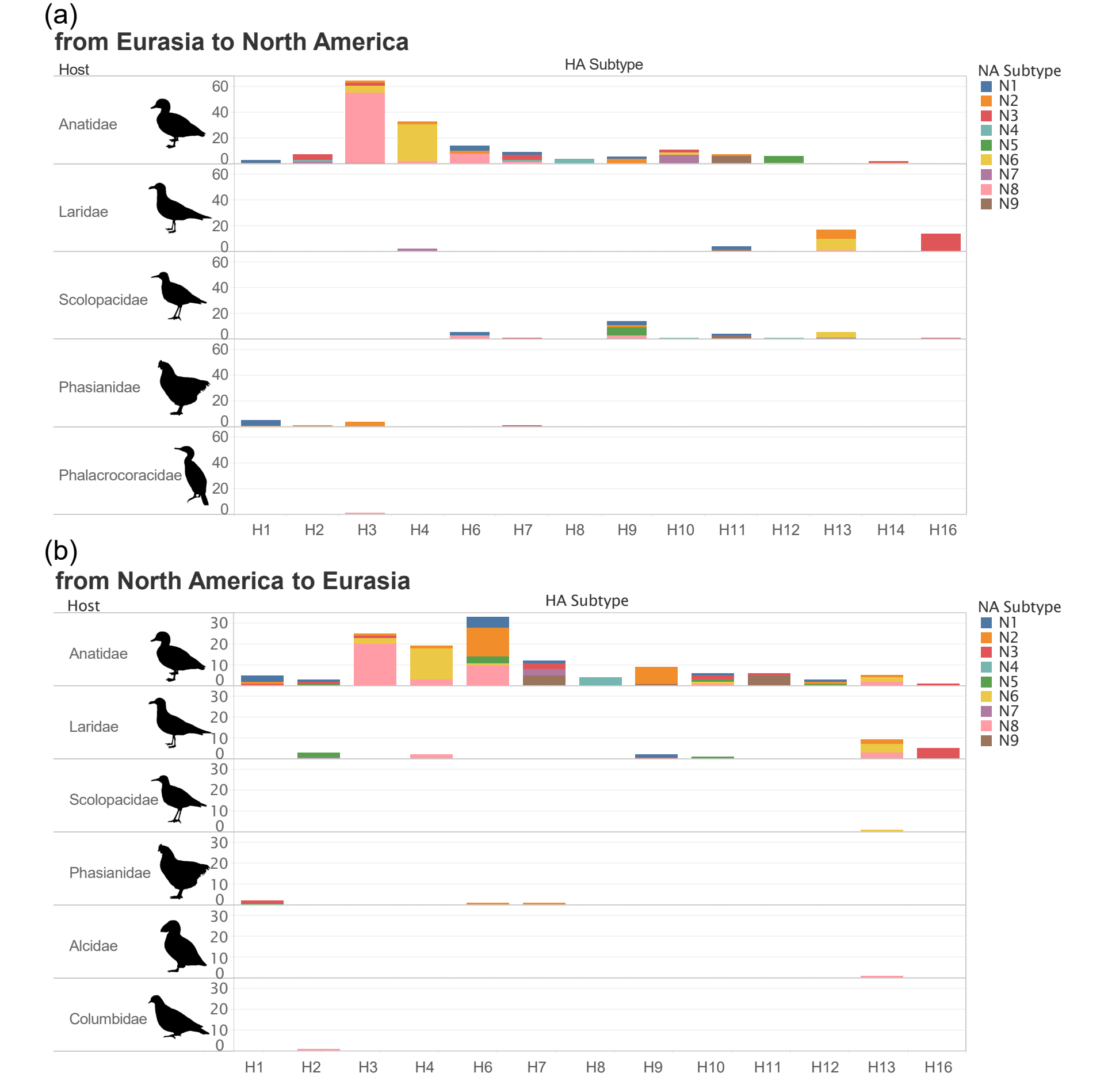


Figure 4. subtypes of intercontinentally transmitted viruses.

Bar graphs show HA subtypes of intercontinental AIVs by host family. The X-axis represents HA subtype, while the Y-axis indicates the number of AIVs. AIVs that lack the NA subtype were excluded. Each NA subtype is represented by a different color. (a) shows transmission from Eurasia to North America, and (b) depicts transmission from North America to Eurasia.

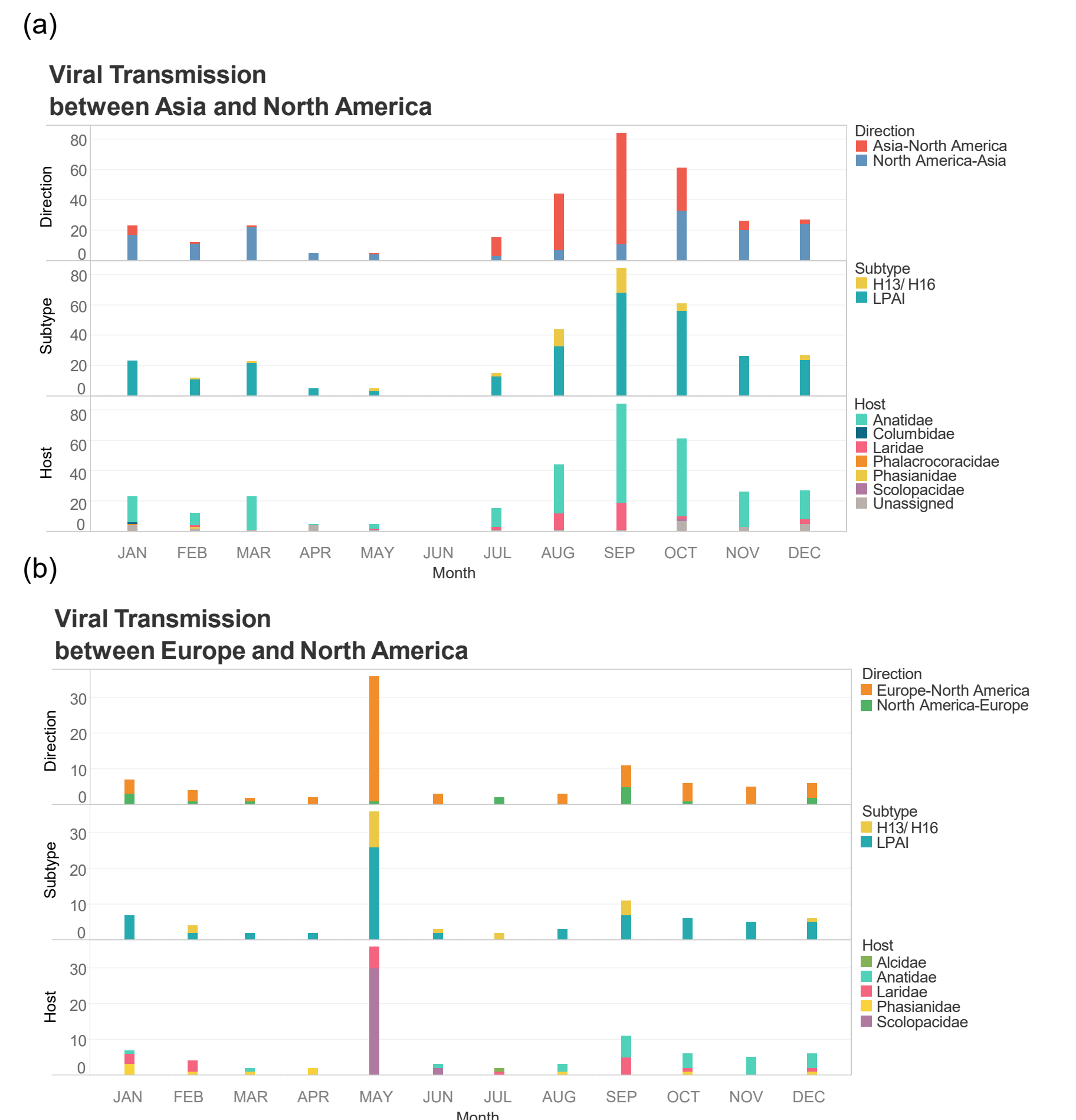


Figure 5. the monthly intercontinental viral transmissions.

Monthly counts of intercontinental AIV transmissions are shown as bar graph with X-axis indicating months and Y-axis showing transmission counts. AIVs lacking month-of-isolation date were excluded. Bars are color-coded by transmission direction, subtype, and host: (a) Viral Transmission between Asia and North America, (b) Viral Transmission between Europe and North America.

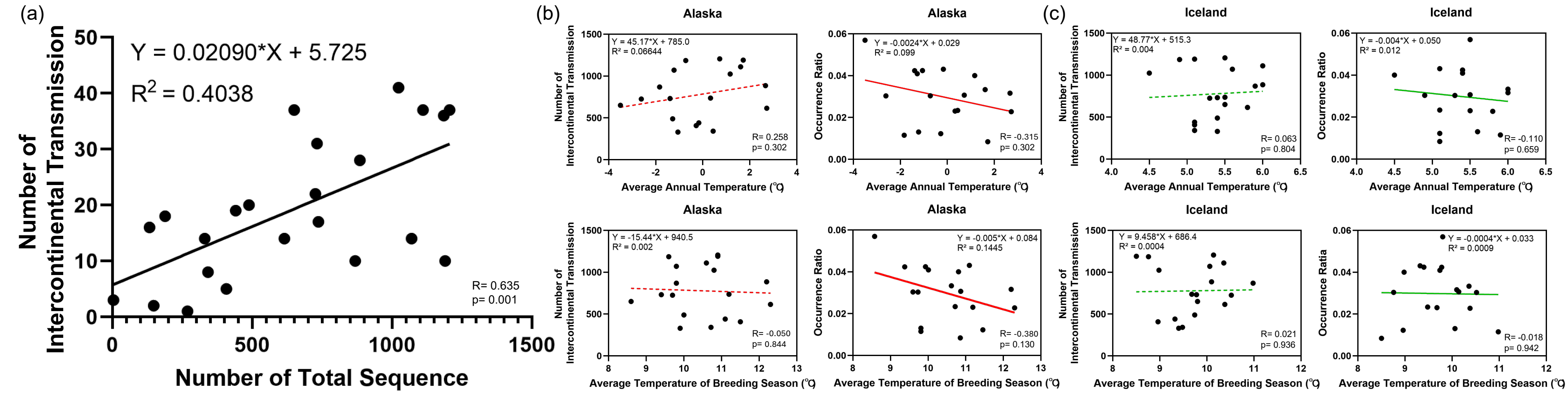


Figure 6. Correlation analysis with factors that can affect the detection of intercontinental viral movement.

(a) Correlation between total number of isolated viruses and the number of isolates. (b) Correlation analysis of annual average temperature and average temperature of breeding season with occurrence ratio and the number of isolates in Alaska. annual average temperature and average temperature of breeding season with occurrence ratio and the number of isolates in Greenland.